File Handling in Java Demonstration

Title: File Handling – Reading and Writing Files in Java and Finding the Frequency of Patterns

Objective / Aim :

To write the given text into a file named "sample.txt", read the file, and find the frequency of the patterns "pe" and "pi".

System Requirements :

Hardware:

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage
- Keyboard and Monitor

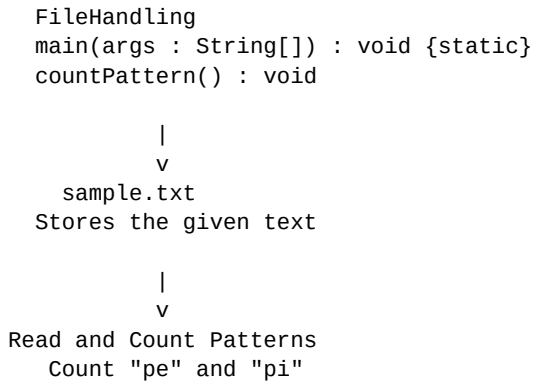
Software:

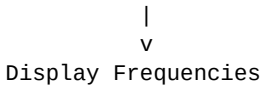
- Operating System: Windows/Linux/macOS
- Java Development Kit (JDK 17 or JDK 21)
- Visual Studio Code / Eclipse IDE or any Java IDE
- Command Prompt/Terminal

Algorithm / Logic Description :

1. Start the program.
2. Create a file named sample.txt.
3. Write the given text into the file using FileWriter.
4. Close the file after writing.
5. Read the contents of the file using FileReader and BufferedReader.
6. Count the occurrences of the patterns "pe" and "pi".
7. Display the frequency of both patterns.
8. Stop the program.

UML Class Diagram :





Relationship: FileHandling interacts with the file system by writing raw text to sample.txt, reading it back sequentially via buffered streams, and computing substring frequencies.

Source Code :

```
import java.io.*;

public class FileHandling {
    public static void main(String[] args) {
        String text = "Peter Piper picked a peck of pickled peppers\\n"
                + "A peck of pickled peppers Peter Piper picked\\n"
                + "If Peter Piper picked a peck of pickled peppers\\n"
                + "Where's the peck of pickled peppers Peter Piper picked?";

        try {
            // Write the text into sample.txt
            FileWriter fw = new FileWriter("sample.txt");
            fw.write(text);
            fw.close();

            // Read the contents of sample.txt
            BufferedReader br = new BufferedReader(
                    new FileReader("sample.txt"));

            StringBuilder content = new StringBuilder();
            String line;

            while ((line = br.readLine()) != null) {
                content.append(line).append("\\n");
            }
            br.close();

            // Count occurrences of pe and pi
            String data = content.toString().toLowerCase();
            int peCount = 0;
            int piCount = 0;

            for (int i = 0; i < data.length() - 1; i++) {
                String pattern = data.substring(i, i + 2);

                if (pattern.equals("pe")) {
                    peCount++;
                }

                if (pattern.equals("pi")) {
                    piCount++;
                }
            }

            System.out.println("'pe' - no of occurrences - "
                    + peCount);
```

```
            System.out.println("'pi' - no of occurrences - "
                    + piCount);

        } catch (IOException e) {
            System.out.println("An error occurred: "
                    + e.getMessage());
        }
    }
}
```

Compilation & Execution Commands :

Save file as: FileHandling.java

Compilation: javac FileHandling.java

Execution: java FileHandling

Output Screenshot :



Result :

Thus, the given text was successfully written into "sample.txt", read using Java file handling operations, and the frequencies of the patterns "pe" and "pi" were counted and displayed.